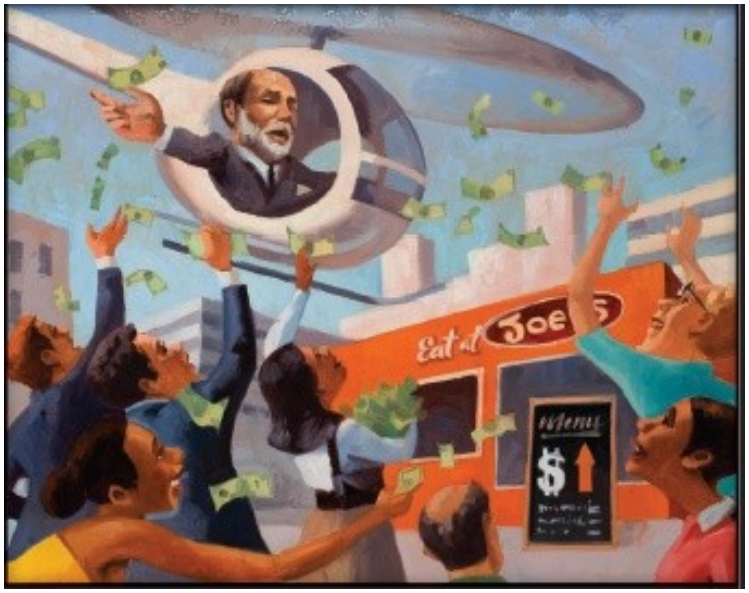
22

Money and Prices in the Long Run



Principles: Chapter 30

PowerPoint Slides prepared by:
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Inflation

- Inflation
  - Increase in the overall level of prices
  - Lower value of money: need more money to buy the same amount of goods
- Deflation
  - Decrease in the overall level of prices
- Hyperinflation
  - Extraordinarily high rate of inflation

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Inflation

- What is inflation?
  - Goods and services becoming more expensive
  - Value of money decreasing ← price of money?
- What would happen if Govt drops money from the sky?

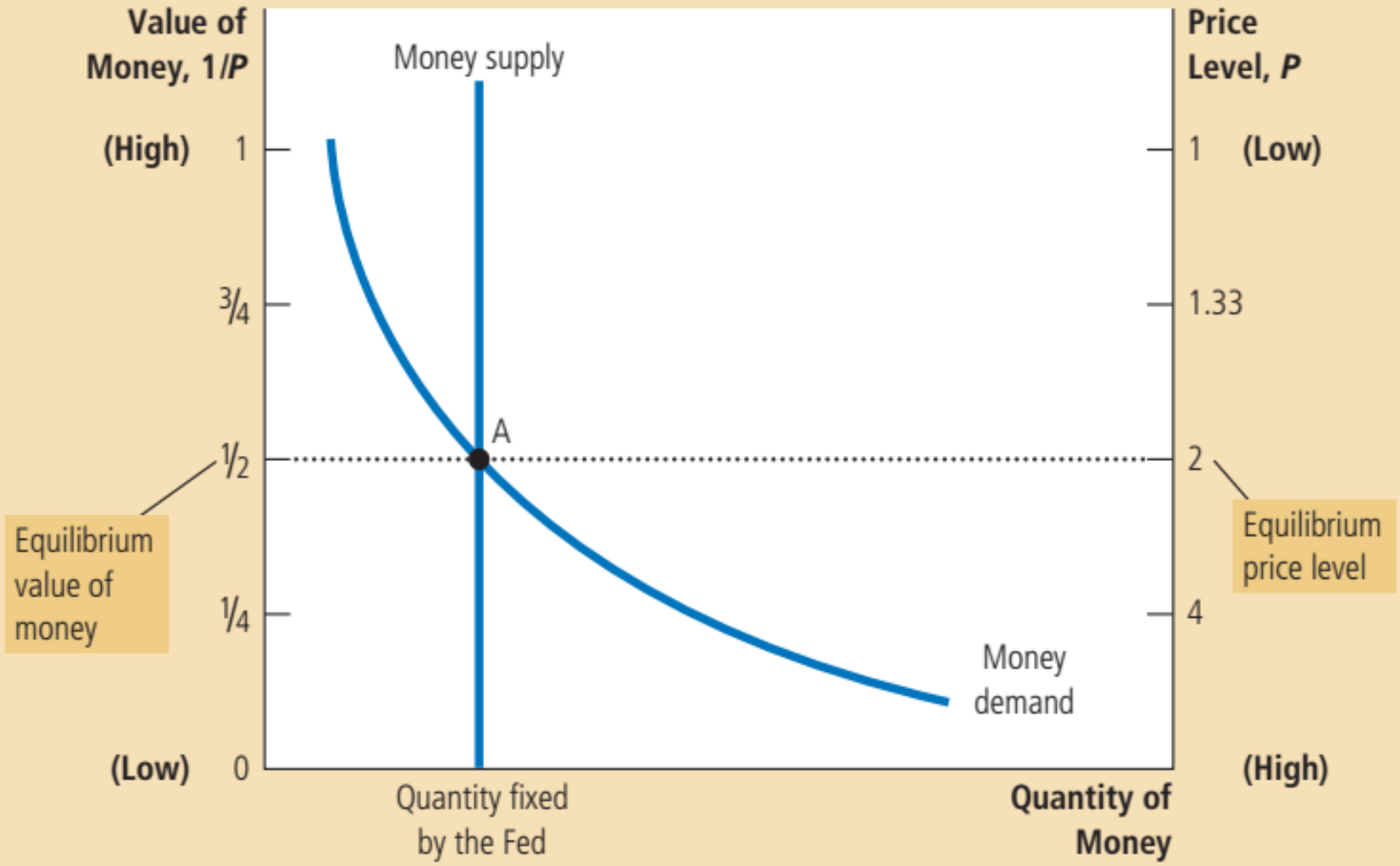
Price level and Value of Money

- If Price level of in a one good economy is 2$, then the value of money is?
- So if P dollars are needed to buy a fixed basket of goods and services, then the quantity that can be bought with 1$ is 1/P.
- Value (price) of money → 1/P.
- As P increases, value of money falls.

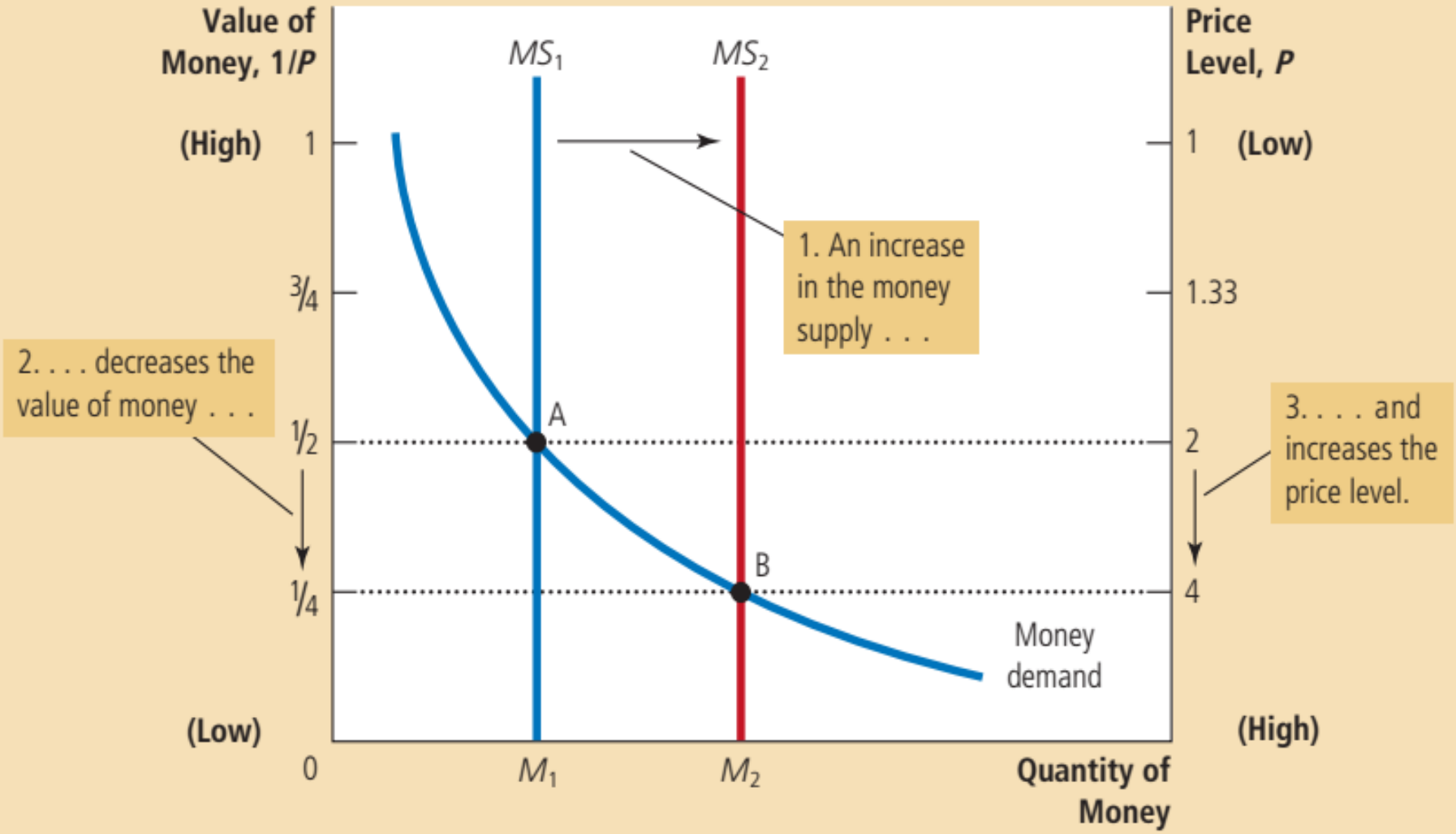
Monetary Equilibrium

- What decides the monetary equilibrium (level of value of money)?
- Money supply decided by?
- Demand for money?
  - Average level of prices in the economy
  - Demand=Supply
  - What if price level is greater than *equilibrium* price level? (Think money as a good)

Monetary Equilibrium



Monetary Injection



Effects of Too Much Money

- What if the Fed raises too much money?
  - Increased supply of money
  - IMMEDIATE Increase in demand of goods and services (short run)
  - Price of goods and services increases
  - Increase in price level, so inflation OVER TIME (long run)

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Effects of a Monetary Injection

- Quantity theory of money
  - The quantity of money available in the economy determines (the value of money) the price level
  - Growth rate in quantity of money available determines the inflation rate

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Classical Dichotomy

- Real Variables: GDP, saving, investment, IR, wage, unemployment
- *Relative prices*
- Nominal Variables: GDP, IR, wage
- Real variables do not depend on money
- *Monetary Neutrality*: Real variables not affected by Money.

Quantity Theory of Money

- $M \times V = P \times Y$
  - P = price level
  - Y = real GDP (quantity of output)
  - $P \times Y$ = dollar value of all goods & services
  - M = quantity of money
  - V = velocity of money (rate at which money changes hands)

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Quantity Theory of Money

Change in money supply, M

- Induces proportional changes in the nominal value of output (P × Y), if V is constant
  - In the long run, Y is determined by real resources (land, labor, capital etc), so the price level (P) increases as MS increases

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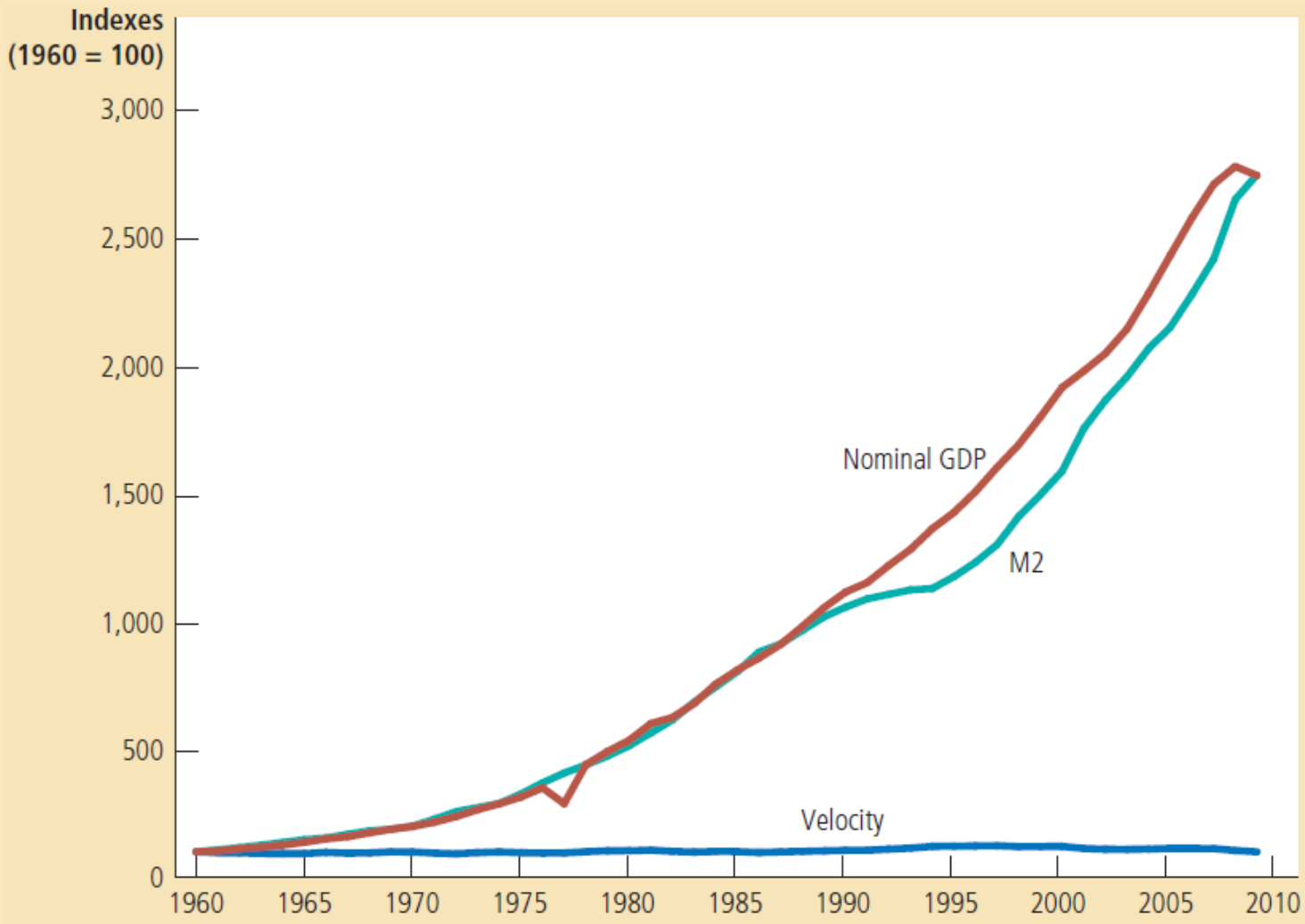
Figure 3

Nominal GDP, the Quantity of Money, and the Velocity of Money



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Money and prices during four hyperinflations

- Four classic hyperinflation, 1920s
  - Austria, Hungary, Germany, and Poland
  - Slope of the money line
    - Rate at which the quantity of money was growing
  - Slope of the price line - inflation rate
  - The steeper the lines - the higher the rates of money growth or inflation
- Prices rise when the government prints too much money

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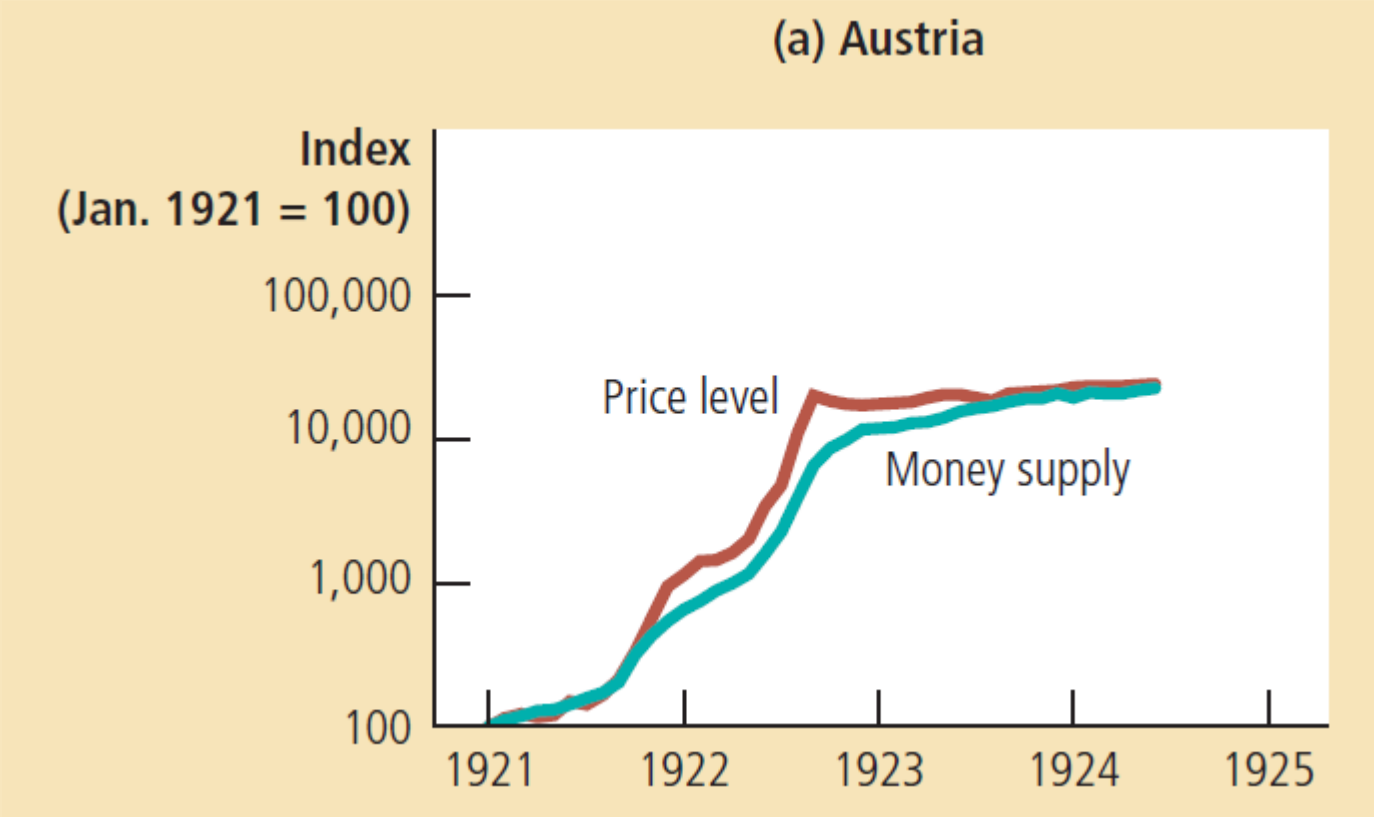
Figure 4

Money and Prices during Four Hyperinflations



This figure shows the quantity of money and the price level during four hyperinflations. (Note that these variables are graphed on logarithmic scales. This means that equal vertical distances on the graph represent equal percentage changes in the variable.) In each case, the quantity of money and the price level move closely together. The strong association between these two variables is consistent with the quantity theory of money, which states that growth in the money supply is the primary cause of inflation.

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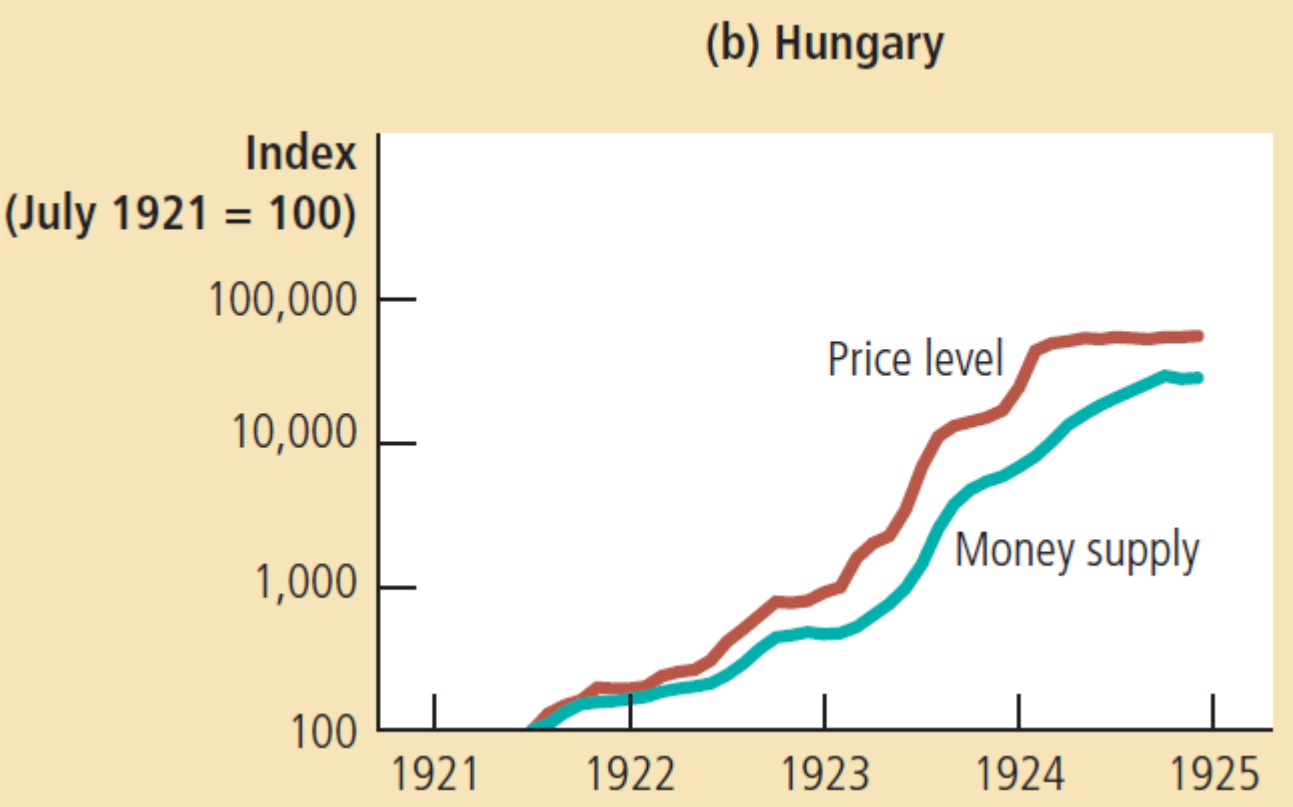
Figure 4

Money and Prices during Four Hyperinflations



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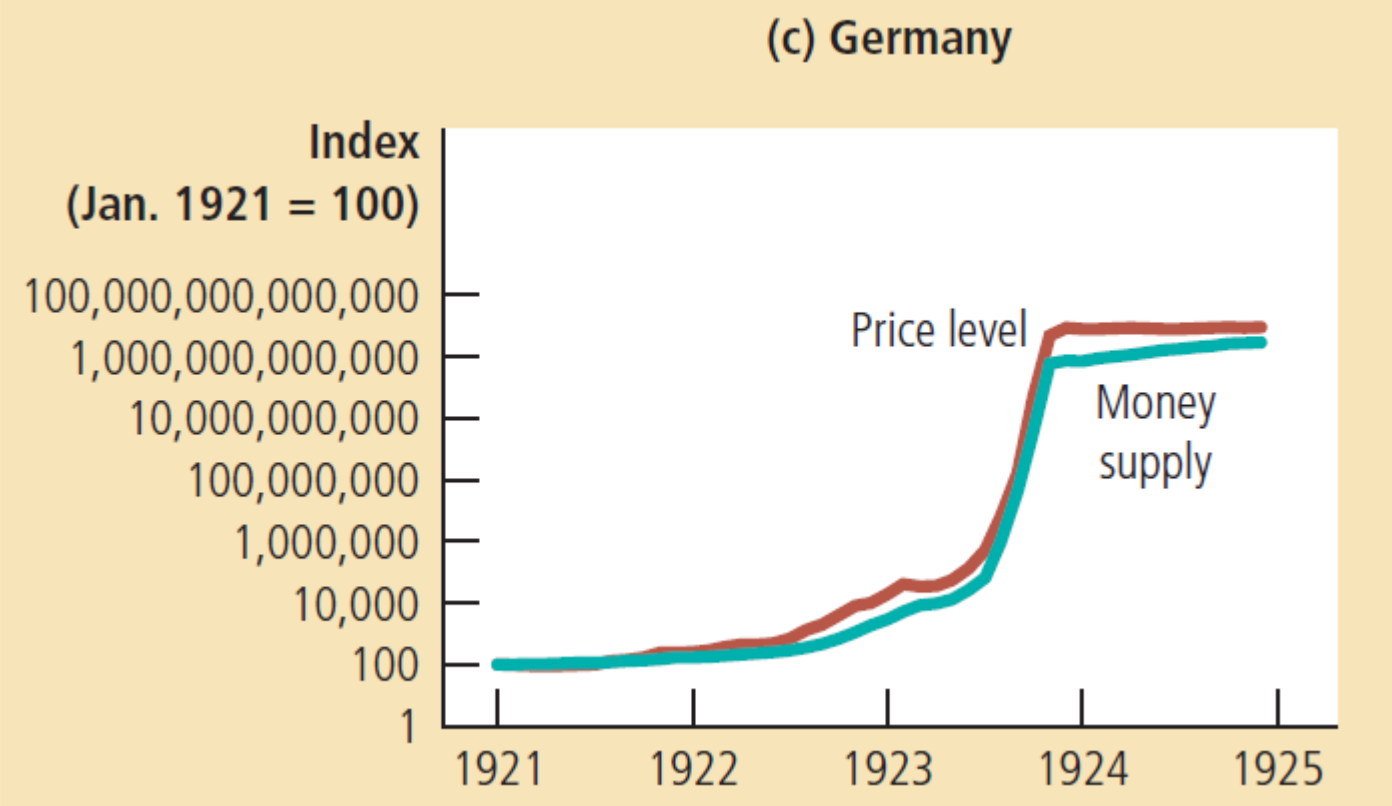

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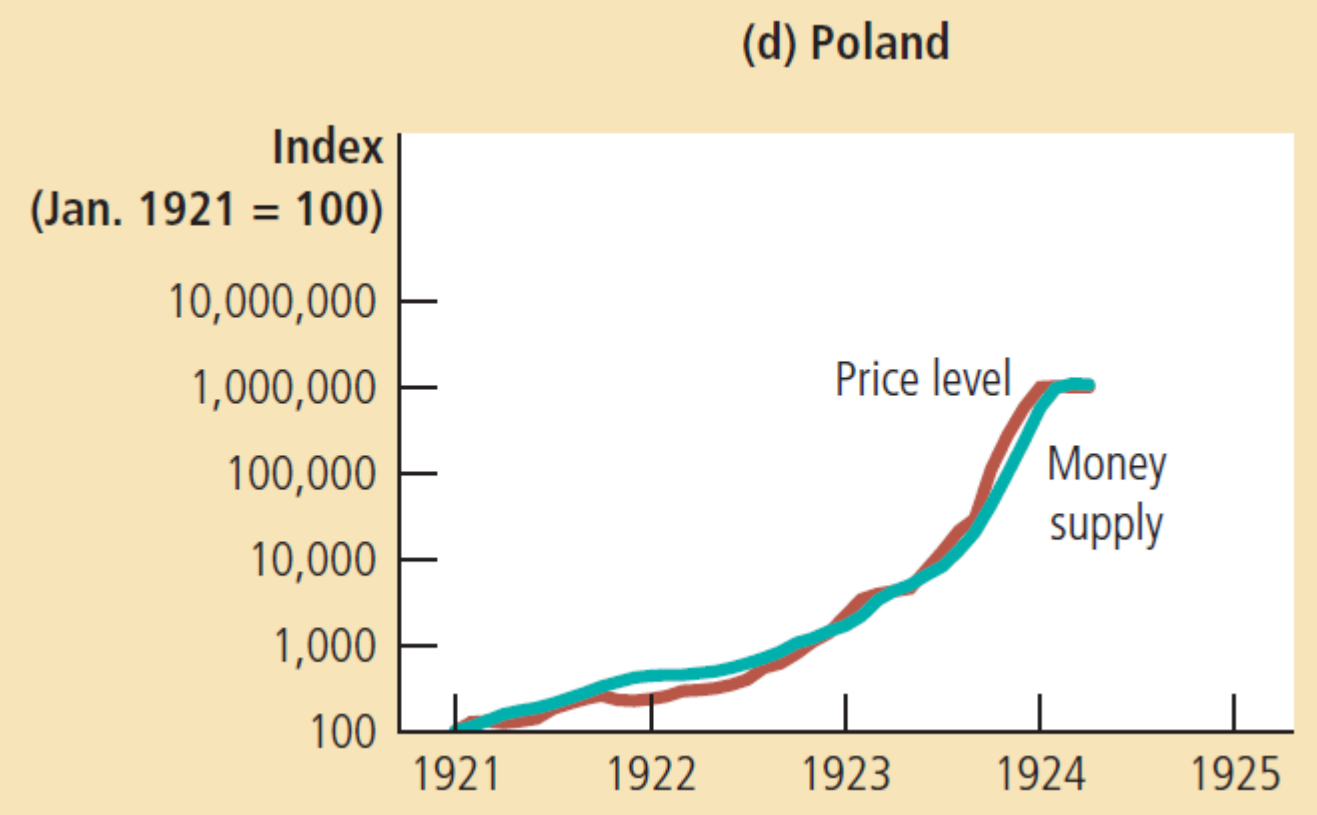

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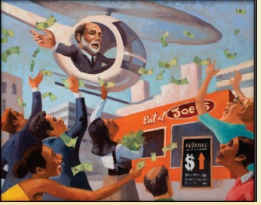
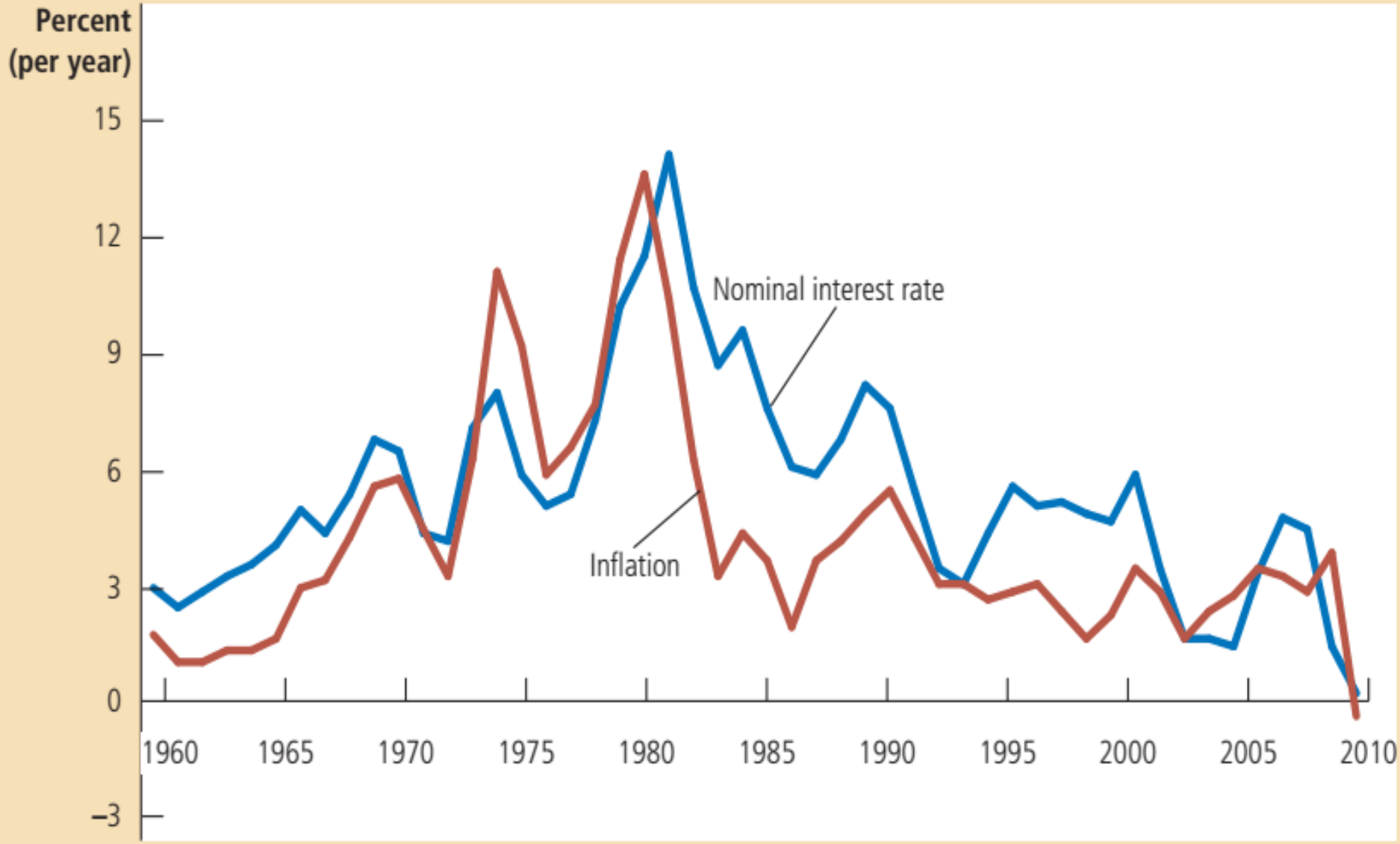
Fisher Effect

- Real IR = Nominal IR - Inflation
- Or NIR = RIR + Inflation
- RIR determined by supply and demand of loanable funds
- Then, higher inflation rate induces increase in NIR, so RIR remains constant
- Short Run vs Long Run




Percent
(per year)
15
12
9
6
3
0
–3
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
Nominal interest rate
Inflation

The Costs of Inflation

- Inflation Tax
- Inflation fallacy
  - "Inflation robs people of the purchasing power of his hard-earned dollars"
- When prices rise
  - Buyers – pay more
  - Sellers – get more
- Inflation does not in itself reduce people's real purchasing power

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The Costs of Inflation

- Shoeleather costs / Menu Costs
- Relative Price variability and misallocation of Resources
- Inflation induced tax distortions
- Arbitary redistributions of wealth
- Inflation reduces saving
  - Inflation reduces the interest income earned
    - Real interest rate = nominal interest rate – inflation rate

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